Analysis plan: GPR153 expression, MGMT co-expression and survival in glioblastoma (revised analysis)

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Summary (approval page)

Status. Exploratory. The authors have seen earlier results on overlapping data. This plan fixes definitions and decision rules before revised values are computed. The manuscript calls it a time-stamped analysis plan, not a preregistration.

Primary hypothesis. In IDH-wildtype primary glioblastoma, GPR153 is the gene with the strongest positive co-expression with MGMT genome-wide.

Primary analysis.

- TCGA-GBM primary tumors, one aliquot per patient, IDH-wildtype: n = 252 (BI 146, GDC-2024 106).
- Batch-adjusted Spearman of MGMT with every eligible gene (TPM >= 1 in >= 20%), from log2(TPM + 1); BH correction. The pooled Spearman is reported with equal prominence.

Pass rule (all four required; branches in 5.6):

1. GPR153 is rank 1 with q < 0.05.
2. GPR153-MGMT is positive with q < 0.05 after adjusting for ESTIMATE score, and separately for a neuronal signature.
3. GPR153-MGMT is positive in the BI-only and the GDC-2024-only subsets (sign check only).
4. GPR153-MGMT adjusted for MGMT promoter methylation, platform and batch is positive with a 95% CI excluding 0.

Expected outcome. The original code's Pearson top-100 MGMT list did not contain GPR153 (listed genes r 0.425-0.548; GPR153 r 0.3645). The authors therefore expect condition 1 to fail and the outcome to be branch (b) or (d).

Primary survival estimate. GPR153 per SD in a Cox model with age and sex (Model 3), unpenalized, in the IDH-wildtype primary cohort; described at most as an age- and sex-adjusted association.

Tiers (Section 15).

- Tier 1: cohort, screen, survival, external co-expression, enrichment, reproducibility deliverables.
- Tier 2: decision-rule adjustments, permutation null, survival replication, multiple imputation, CAMERA, checklists; optional MGMT-ranked enrichment.
- Tier 3: differential expression, survival forest.

Drop order: MGMT-ranked enrichment, DE, forest, multiple imputation, CAMERA. Each dropped analysis is logged in DEVIATIONS.md. Scope rule and independent reproduction rule in 13.2.

Seen before this plan (Section 1):

- submitted manuscript values
- re-runs of the original scripts on the original 391 aliquots
- earlier external values (GSE13041, GSE4290, CGGA), including same-session all-platform and GPL570 GSE13041 values and an all-histology GSE4290 value
- earlier enrichment
- the original code's own Pearson top-100 MGMT list (GPR153 not in it)
- clinical and molecular annotations
- sample-pair duplicate checks
- three raw MGMT intensities in GSE16011 (excluded)

The original-code re-runs computed GPR153-MGMT statistics on all 391 aliquots, which include the revised cohort's samples. No value has been computed by the revised pipeline.